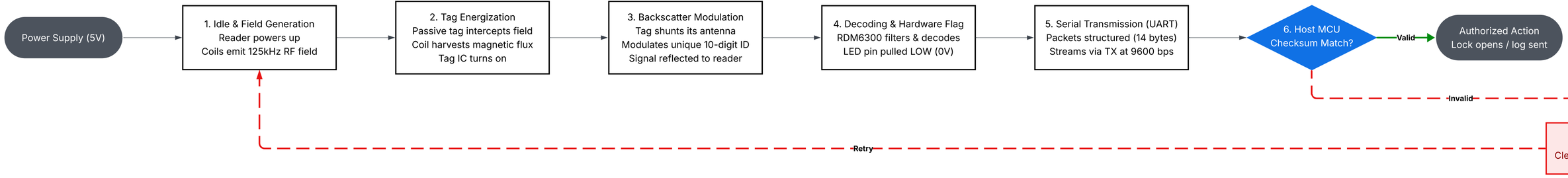
Power Supply (5V)
1. Idle & Field Generation
Reader powers up
Coils emit 125kHz RF field
2. Tag Energization
Passive tag intercepts field
Coil harvests magnetic flux
Tag IC turns on
3. Backscatter Modulation
Tag shunts its antenna
Modulates unique 10-digit ID
Signal reflected to reader
4. Decoding & Hardware Flag
RDM6300 filters & decodes
LED pin pulled LOW (0V)
5. Serial Transmission (UART)
Packets structured (14 bytes)
Streams via TX at 9600 bps
6. Host MCU
Checksum Match?
Valid
Authorized Action
Lock opens / log sent
Invalid
Discard Frame
Clear serial buffer & ignore
Retry